

If a judge moves a slider, here is why it missed.

Every number below is the **median of 150 drops** from the same model the page runs, one control moved at a time, everything else at the page default (wind 1.6 m/s, direction 45°, height 40 m, weight 1.2 kg/m², compass +6°, works the wind out). The 90th percentile is the unlucky drop.

The sentence that covers almost everything

“The path planner can only fly paths the canopy can physically fly. In the last seconds the finest correction available is the turn radius — so if you make it heavier, you make it turn wider, and the miss follows.”

Weight per m² of canopy slider 0.25 – 2.5 · the important one

WEIGHT	FLIES	TURN RADIUS	MEDIAN	90TH
0.25	2.1 m/s	1.2 m	0.4 m	1.0 m
0.50	2.9 m/s	2.4 m	1.5 m	4.5 m
0.75	3.6 m/s	3.6 m	4.3 m	6.9 m
1.20	4.5 m/s	5.7 m	4.3 m	10.4 m
1.50	5.1 m/s	7.2 m	8.0 m	13.0 m
2.50	6.5 m/s	11.9 m	7.5 m	15.1 m

The median wobbles; **the 90th percentile does not** — it rises without exception, 1.0 → 4.5 → 6.9 → 9.1 → 10.4 → 13.0 → 14.2 → 15.1 m. Quote that one. Heavier does not just miss by more on average, it makes the bad drops much worse.

Say this

“That is the finding, and it is panel 06. Turn radius rises with weight, so a heavier canopy cannot make fine corrections at the end. The right weight is the lightest one that can still fly into the wind.”

Wind speed slider 0 – 4 m/s

WIND	MEDIAN	90TH	
0.0	1.7 m	5.9 m	nothing to fight
1.6	4.3 m	10.4 m	the default
2.5	8.5 m	10.8 m	worst on this heading
4.0	3.3 m	8.7 m	improves — see below

Careful – this one is not a straight line

The miss gets worse to about 2.5 m/s and then *improves*. That is not a bug and you should not be caught out by it: at the default direction of 45° the wind blows roughly **towards** the target, so a strong wind carries the package to where it was going anyway. Move the direction slider to 180° and the miss climbs again. **Speed alone does not decide it; speed and direction together do.**

Wind direction slider 0 – 360°

DIRECTION	MEDIAN	90TH	
90° / 315°	1.6 m	2.9 m	best
45°	4.3 m	10.4 m	the default
180°	6.9 m	8.1 m	worst — headwind on approach

Drop height slider 20 – 60 m

HEIGHT	TIME ALOFT	MEDIAN	90TH
20 m	13 s	2.7 m	8.1 m
40 m	26 s	4.3 m	10.4 m
60 m	39 s	6.5 m	11.0 m

If asked why higher is worse

“More height means more time in the wind, so more total drift to undo — and the compass error has longer to accumulate. The extra time to correct does not pay for the extra drift.”

How wrong the compass is slider -12 – +12°

COMPASS ERROR	MEDIAN	90TH
0°	1.3 m	5.7 m
+6° (default)	4.3 m	10.4 m
-12°	5.7 m	7.5 m

Roughly symmetric, worse at both extremes. The filter estimates this bias as a fourth unknown, which is why it degrades gracefully rather than breaking.

What it knows about the wind

SETTING	MEDIAN	90TH
Nothing at all	23.8 m	25.3 m
Works it out (this project)	4.3 m	10.4 m
Simply handed the answer	1.1 m	2.2 m

Trap – be ready for this one

At the page's default settings, **being handed the true wind clearly beats working it out**: your method is closer in only 18 drops out of 150. Do not be surprised by it and do not claim otherwise.

Say: “That third option is cheating — it is the best case, and it is there as the yardstick. My claim is not that working the wind out beats being told it; the paper shows those two are not separated at this sample size. My claim is that working it out beats *not knowing* — 23.8 m down to 4.3 m — and nothing on board has to measure the wind to do it.”

Force it to fly straight section 03

SETTING	MEDIAN	90TH
Forced straight	129.9 m	129.9 m
Free to turn	4.3 m	10.4 m

Why both numbers are identical

With steering disabled the heading never changes, so every drop flies the same line and lands in the same place — there is nothing left for chance to affect. It is not a coincidence and it is not a bug.

Say: “That switch turns the steering off, so it just flies until it lands. It is there to show the observability point: on a fixed heading a headwind and a slow airspeed look identical, so the filter cannot separate them. Turning is what makes the wind measurable.”

The one to steer a judge towards

Set **weight to 0.25** and leave everything else alone: **0.4 m median, 1.0 m at the 90th percentile**, turn radius 1.2 m. Then drag weight to 2.5 and drop again. That single slider, moved twice, is the whole project.

If a number on screen does not match this sheet

A single drop is one sample, and these are medians of 150. The page shows you one drop at a time; press *50 drops at once* to compare like for like. Never defend a single drop — defend the distribution.